

Design and Implementation of the Lenskart Eyewear Retail Database

CHAPTER 1

1 Introduction:

The Lenskart Database is designed to represent the data structure of an online eyewear retail platform. Lenskart is one of India's leading eyewear companies, offering eyeglasses, sunglasses, contact lenses, and related accessories through its online platform and physical stores. Managing such a large volume of products, customers, orders, payments, and inventory requires a well-organized relational database.

This project focuses on designing a structured database that stores and manages the essential information involved in Lenskart's business operations. The database includes entities such as Customers, Products, Categories, Brands, Orders, Order Details, Payments, Inventory, Reviews, and Wishlist. These entities are connected through appropriate relationships using primary keys and foreign keys to ensure data consistency and integrity.

The database is designed using Oracle SQL by applying relational database concepts such as entity identification, relationship mapping, normalization, constraints, and SQL operations. The purpose of this database is to organize business data efficiently, reduce redundancy, and enable accurate storage, retrieval, and management of information. This project demonstrates how a real-world e-commerce database can be structured to support daily business activities while maintaining data accuracy and reliability.

1.1 Problem Statement:

Lenskart handles a large volume of business data, including customer information, product details, orders, payments, inventory, and customer reviews. Managing this data manually or through unorganized systems can lead to data redundancy, inconsistencies, slow data retrieval, and difficulties in maintaining accurate business records.

A well-designed relational database is required to organize and manage this information efficiently. The database should establish proper relationships between different entities, maintain data integrity, reduce redundancy through normalization, and enable quick retrieval of information for daily business operations. This project focuses on designing a structured Oracle SQL database that supports the core activities of an online eyewear retail platform while ensuring efficient data storage and management.

1.3 Objectives:

- To design a relational database based on the business operations of Lenskart.

- To identify the major entities involved in the database, such as Customer, Product, Category, Brand, Orders, Order Details, Payment, Inventory, Reviews, and Wishlist.
- To establish relationships between the entities using primary keys and foreign keys.
- To organize data efficiently by applying normalization techniques to minimize redundancy.
- To create database tables with appropriate constraints such as PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, and CHECK constraints.
- To ensure data accuracy, consistency, and integrity within the database.
- To perform SQL operations for storing, retrieving, updating, and managing business data.
- To provide a database structure that supports the efficient management of customer, product, and order information.
- To gain practical experience in designing and implementing relational databases using Oracle SQL.

1.4 Motive:

The main motive of this project is to design a well-structured relational database based on the business operations of Lenskart. As an online eyewear retailer, Lenskart manages a large amount of data related to customers, products, orders, inventory, and payments. A properly designed database is essential for organizing this information efficiently and ensuring smooth business operations.

This project aims to apply database design principles using Oracle SQL by creating related tables, defining relationships, and maintaining data integrity through appropriate constraints. It also provides practical knowledge of how a real-world e-commerce database is structured to support accurate data storage, efficient retrieval, and effective management of business information.

1.5 Stakeholders:

The stakeholders involved in the Lenskart database include:

- **Customers:** Purchase eyewear products, manage their accounts, place orders, make payments, and provide product reviews.
- **Database Administrator (DBA):** Maintains the database, ensures data security, performs backups, and manages database performance.

- **Store Administrator:** Manages product information, inventory, pricing, categories, and order records.
- **Sales and Customer Support Team:** Access customer and order information to resolve customer queries and provide support.
- **Inventory Manager:** Monitors product availability, stock levels, and inventory updates across stores and warehouses.
- **Management Team:** Uses reports and business data to analyze sales performance, customer trends, and inventory status for decision-making.

1.6 Business Requirements:

The database should support the core business operations of an online eyewear retail platform by providing efficient storage and management of business information.

The database should:

- Maintain customer information securely.
- Store detailed information about eyewear products, categories, brands, and prices.
- Manage customer orders and order details.
- Record payment information for completed transactions.
- Maintain inventory and product availability.
- Allow customers to add products to their wishlist.
- Store customer ratings and product reviews.
- Support accurate reporting of products, orders, sales, and inventory.
- Ensure data consistency and minimize duplication through proper relational design.

1.7 Functional Requirements:

The database should be capable of performing the following functions:

- Store customer registration and profile information.
- Store product details including category, brand, specifications, and price.
- Manage customer orders and order history.
- Record multiple products within a single order.
- Store payment details for each completed order.

- Track product inventory and stock availability.
- Store customer reviews and ratings for products.
- Maintain customer wishlists.
- Retrieve information efficiently using SQL queries.
- Support CRUD (Create, Read, Update, Delete) operations for all database entities.
- Maintain relationships between tables using primary keys and foreign keys.

1.8 Non-Functional Requirements:

Security

- Customer information and payment records should be protected from unauthorized access.
- Only authorized users should be allowed to modify sensitive database records.
- Data integrity should be maintained using database constraints such as Primary Keys, Foreign Keys, UNIQUE, NOT NULL, and CHECK constraints.

Reliability

- The database should provide accurate and consistent information during all operations.
- Data should remain reliable even when multiple transactions occur.
- Backup and recovery mechanisms should be available to prevent data loss.

Performance

- The database should retrieve customer, product, and order information efficiently.
- SQL queries should execute within an acceptable response time even when handling large volumes of data.

Scalability

- The database should support future expansion, including additional products, customers, stores, and business operations without requiring major structural changes.

Availability

- The database should be available whenever required to support daily business activities and customer transactions.

Maintainability

- The database structure should be easy to maintain, update, and modify as business requirements evolve.
- Proper normalization should reduce redundancy and simplify future maintenance.

1.9 Conclusion:

The Lenskart database project demonstrates the design of a structured and efficient relational database using Oracle SQL. It organizes essential business data such as customers, products, orders, payments, inventory, and reviews while maintaining data integrity and reducing redundancy. This project provides practical knowledge of database design concepts and highlights the importance of a well-organized database in supporting the operations of an online eyewear retail platform.